exponential family $\mathcal{E}$ with cumulant function F is given by:

$$g_{ij}(\theta) = \partial_i \partial_j F(\theta), \tag{190}$$

$$\Gamma^{\alpha}_{ij,k} = \frac{1-\alpha}{2} \partial_i \partial_j \partial_k F(\theta). \tag{191}$$

When $\alpha = 1$, $\Gamma^{\alpha}_{ij,k} = 0$ and ∇^1 is flat, and so is ∇^{-1} by using the fundamental theorem of information geometry.

The ± 1-structure can also be derived from a *divergence manifold structure* by choosing the reverse Kullback-Leibler divergence KL^*:

$$(\mathcal{E}, {}_{\mathcal{E}}g, {}_{\mathcal{E}}\nabla^e, {}_{\mathcal{E}}\nabla^m) \equiv (\mathcal{E}, \mathrm{KL}^*). \tag{192}$$

Therefore, the Kullback-Leibler divergence $\mathrm{KL}[P_\theta : P_{\theta'}]$ amounts to a Bregman divergence (for the cumulant function of the exponential family):

$$\mathrm{KL}^*[P_{\theta'} : P_\theta] = \mathrm{KL}[P_\theta : P_{\theta'}] = B_F(\theta' : \theta). \tag{193}$$

The *best exponent error* α^* of the best *Maximum A Priori* (MAP) decision rule is found by minimizing the *Bhattacharyya distance* to get the *Chernoff information* [106]:

$$C[P_1, P_2] = -\log \min_{\alpha \in (0,1)} \int_{x \in \mathcal{X}} p_1^{\alpha}(x) p_2^{1-\alpha}(x) \mathrm{d}\mu(x) \geq 0. \tag{194}$$

On the exponential family manifold $\mathcal{E}$, the *Bhattacharyya distance*:

$$B_\alpha[p_1 : p_2] = -\log \int_{x \in \mathcal{X}} p_1^{\alpha}(x) p_2^{1-\alpha}(x) \mathrm{d}\mu(x), \tag{195}$$

amounts to a *skew Jensen parameter divergence* [83] (also called Burbea-Rao divergence):

$$J_F^{\alpha}(\theta_1 : \theta_2) = \alpha F(\theta_1) + (1-\alpha) F(\theta_2) - F(\theta_1 + (1-\alpha)\theta_2). \tag{196}$$

It can be shown that the Chernoff information (that minimizes α) is equivalent to a Bregman divergence: Namely, the Bregman divergence for exponential families at the optimal exponent value α^*.

Theorem 10 (Chernoff information [73])**.** *The Chernoff information between two distributions belonging to the same exponential family amount to a Bregman divergence:*

$$C[P_{\theta_1} : P_{\theta_2}] = B(\theta_1 : \theta_{12}^{\alpha^*}) = B(\theta_2 : \theta_{12}^{\alpha^*}), \tag{197}$$

where $\theta_{12}^{\alpha} = (1-\alpha)\theta_1 + \alpha\theta_2$, and α^* denote the best exponent error.

Let $\theta_{12}^* := \theta_{12}^{\alpha^*}$ denote the best exponent error. The geometry [73] of the best error exponent can be explained on the dually flat exponential family manifold as follows:

$$P^* = P_{\theta_{12}^*} = G_e(P_1, P_2) \cap \mathrm{Bi}_m(P_1, P_2), \tag{198}$$

where G_e denotes the exponential geodesic γ_{∇^e} and Bi_m the m-bisector:

$$\mathrm{Bi}_m(P_1, P_2) = \{P \ : \ F(\theta_1) - F(\theta_2) + \eta(P)^\top (\theta_2 - \theta_1) = 0\}. \tag{199}$$

Figure 12 illustrates how to retrieve the best error exponent from an exponential arc (θ-geodesic) intersecting the m-bisector.

Furthermore, instead of considering two distributions for this statistical binary decision problem, we may consider a set of n distributions of $P_1, \ldots, P_n \in \mathcal{E}$. The geometry of the error exponent in this multiple hypothesis testing setting has been investigated in [72]. On the dually flat exponential family manifold, it corresponds to check the exponential arcs between *natural neighbors* (sharing Voronoi subfaces) of a Bregman Voronoi diagram [20]. See Figure 13 for an illustration.

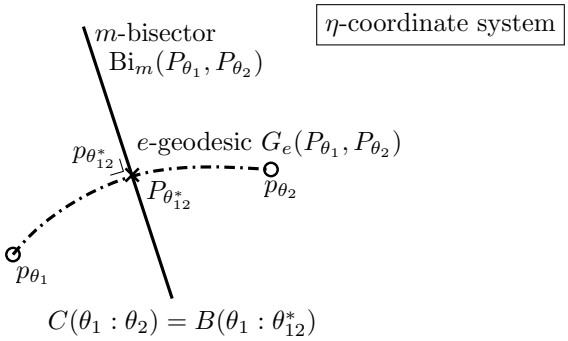

Figure 12: Exact geometric characterization (not necessarily i closed-form) of the best exponent error rate α^*.

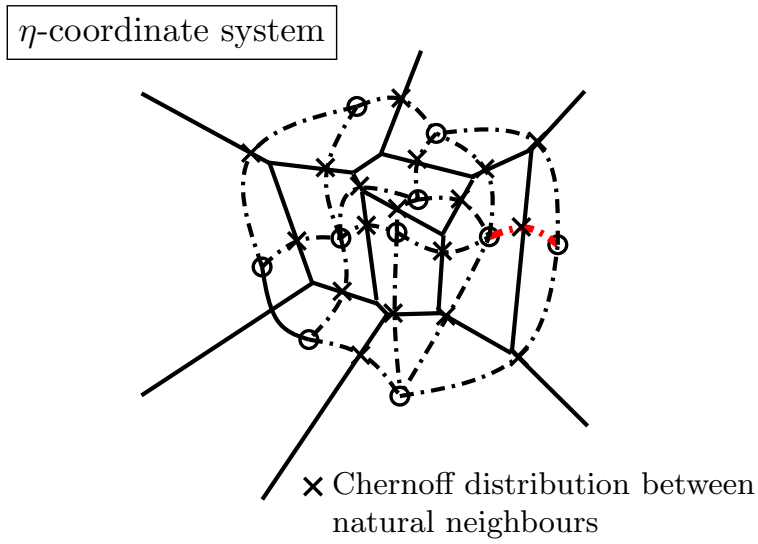

Figure 13: Geometric characterization of the best exponent error rate in the multiple hypothesis testing case.

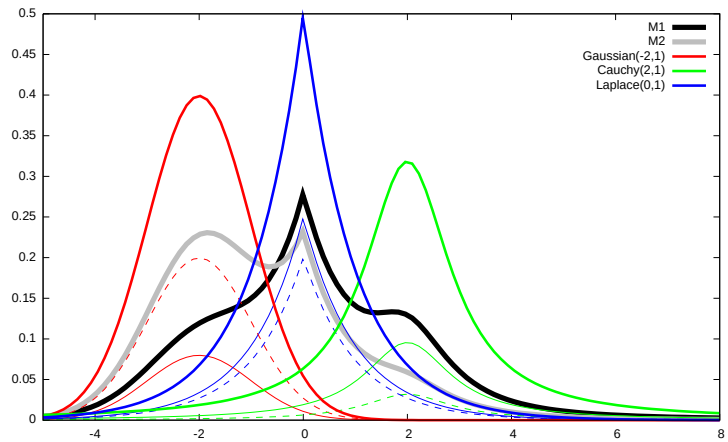

Figure 14: Example of a mixture family of order $D = 2$ (3 components: Laplacian, Gaussian and Cauchy prefixed distributions).

4.4 Clustering mixtures in the dually flat mixture family manifold $(\mathcal{M}, \mathrm{KL})$

Given a set of k prescribed statistical distributions $p_0(x), \ldots, p_{k-1}(x)$, all sharing the same support $\mathcal{X}$ (say, $\mathbb{R}$), a *mixture family* $\mathcal{M}$ of order $D = k - 1$ consists of all *strictly convex combinations* of these component distributions [93]:

$$\mathcal{M} := \left\{ m(x;\theta) = \sum_{i=1}^{k-1} \theta_i p_i(x) + \left(1 - \sum_{i=1}^{k-1} \theta_i\right) p_0(x) \text{ such that } \theta_i > 0, \sum_{i=1}^{k-1} \theta_i < 1 \right\}. \tag{200}$$

Figure 14 displays two mixtures obtained as convex combinations of prescribed Laplacian, Gaussian and Cauchy component distributions ($D = 2$). When considering a set of prescribed Gaussian component distributions, we obtain a w-Gaussian Mixture Model, or w-GMM for short.

We consider the expected information manifold $(\mathcal{M}, {}^{\mathcal{M}}g, {}^{\mathcal{M}}\nabla^m, {}^{\mathcal{M}}\nabla^e)$ which is dually flat and equivalent to (M_Θ, KL). That is, the KL between two mixtures with prescribed components (w-mixtures, for short) is equivalent to a Bregman divergence for $F(\theta) = -h(m_\theta)$, where $h(p) = \int p(x) \log p(x) \mathrm{d}\mu(x)$ is the differential Shannon information (negative entropy) [93]:

$$\mathrm{KL}[m_{\theta_1} : m_{\theta_2}] = B_F(\theta_1 : \theta_2). \tag{201}$$

Consider a set $\{m_{\theta_1}, \ldots, m_{\theta_n}\}$ of n w-mixtures [93]. Because $F(\theta) = -h(m(x;\theta))$ is the *negative differential entropy* of a mixture (not available in closed form [95]), we approximate the untractable F by another close tractable generator $\tilde{F}$. We use Monte Carlo stochastic sampling to get Monte-Carlo convex $\tilde{F}_{\mathcal{S}}$ for an independent and identically distributed sample $\mathcal{S}$.

Thus we can build a *nested sequence* $(\mathcal{M}, \tilde{F}_{\mathcal{S}_1}), \ldots, (\mathcal{M}, \tilde{F}_{\mathcal{S}_m})$ of tractable dually flat manifolds for nested sample sets $\mathcal{S}_1 \subset \ldots \subset \mathcal{S}_m$ converging to the ideal mixture manifold $(\mathcal{M}, F)$: $\lim_{m\to\infty}(\mathcal{M}, \tilde{F}_{\mathcal{S}_m}) = (\mathcal{M}, F)$ (where convergence is defined with respect to the induced canonical Bregman divergence). A key advantage of this approach is that for a given sample $\mathcal{S}$, all computations carried inside the dually flat manifold $(\mathcal{M}, \tilde{F}_{\mathcal{S}})$ are *consistent*, see [93].

For example, we can apply Bregman k-means [87] on these Monte Carlo dually flat spaces [85] of w-GMMs (Gaussian Mixture Models) to cluster a set of w-GMMs. Figure 15 displays the result of such a clustering.

We have briefly described two applications using dually flat manifolds: (1) the *dually flat exponential manifold* induced by the statistical reverse Kullback-Leibler divergence on an exponential family (structure $(\mathcal{E}, \mathrm{KL}^*)$), and (2) the *dually flat mixture manifold* induced by the statistical Kullback-Leibler divergence